Exercise 1:

(a) Assumptions:
(1) Customer Number is a unique identifier for a customer.
(2) One order form represents one transaction at a specific date and time, and an order is uniquely identified by the combination of Customer Number, Date, and Time.
(3) An employee may assist many customers, and a customer may place many orders.
(4) Part Number is a unique identifier for a specific part.
(5) Cage Code identifies a physical storage location, and a part exists in only one cage.

(b) Normalizing Process:
(1) Step 1: 1NF
We flatten the form into a single table. The primary key must identify a specific line item on a specific order.

customerNumber	orderDate	orderTime	customerName	customerType	employeeName	partNumber	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	7/1/2024	10:30 am	Jeff Peterson	Consumer	D. Harrison	10654	Float Control	Plumbing	G413	4	12
HG54587	7/1/2024	10:30 am	Jeff Peterson	Consumer	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	7/1/2024	10:30 am	Jeff Peterson	Consumer	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	7/1/2024	10:30 am	Jeff Peterson	Consumer	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

The primary key (PK) is this table can be a composite key consisting of **customerNumber, orderDate, orderTime, and partNumber**.
This is 1NF because the data do not have repeating groups, and all of them are atomic values.

(2) Step 2: 2NF
A table is in 2NF if it is already in 1NF and there are no partial dependencies, meaning every non-key attribute must be fully functionally dependent on the entire primary key, not just part of it. In the above table, several attributes depend only on a part of the composite key. For examples, customerName and customerType depend only on customerNumber, while partName, partType, cageCode, and unitPrice depend only on partNumber. In addition, employee only depends on the customerNumber, orderDate, and orderTime, rather than the full composite key including partNumber. Because these non-key fields are determined by only part of the primary key, the table contains partial dependencies, so it is not in 2NF and must be decomposed into separate relations that align attributes with what they depend on.

customer

customerNumber	customerName	customerType
HG54587	Jeff Peterson	Consumer

part

partNumber	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order

customerNumber	**orderDate**	**orderTime**	employee
HG54587	7/1/2024	10:30 am	D. Harrison

Primary Key: customerNumber, orderDate, orderTime

order_line

customerNumber	**orderDate**	**orderTime**	**partNumber**	quantityOrdered
HG54587	7/1/2024	10:30 am	10654	4
HG54587	7/1/2024	10:30 am	10456	3
HG54587	7/1/2024	10:30 am	10776	7
HG54587	7/1/2024	10:30 am	10657	5

Primary Key: customerNumber, orderDate, orderTime, partNumber

Step 3: 3NF
A table is in 3NF if it is already in 1NF and 2NF and there are no transitive dependencies, meaning that no non-key attribute depends on another non-key attribute instead of directly on the primary key. For 2NF above, in the customer table, customerName and customerType depend directly on customerNumber. In the order table, employee depends directly on the order's composite primary key (customerNumber, orderDate, orderTime). In the order_line table, quantityOrdered depends on the full composite key (customerNumber, orderDate, orderTime, partNumber). However, the part table is not in 3NF because it contains a transitive dependency, where partNumber determines partName, and partName determines partType. Thus, partType depends on the primary key indirectly, and we need to split the part table into new tables to reach 3NF.

customer

customerNumber	customerName	customerType
HG54587	Jeff Peterson	Consumer

part

partNumber	partName	cageCode	unitPrice
10654	Float Control	G413	12
10456	Modulator	H433	7
10776	Hose Assembly	G413	9
10657	Float Assembly	G413	10

partName	partType
Float Control	Plumbing
Modulator	Electrical
Hose Assembly	Plumbing
Float Assembly	Plumbing

order

customerNumber	**orderDate**	**orderTime**	employee
HG54587	7/1/2024	10:30 am	D. Harrison

Primary Key = **(customerNumber, orderDate, orderTime)**
Foreign Key = *customerNumber*

order_line

customerNumber	**orderDate**	**orderTime**	***partNumber***	quantityOrdered
HG54587	7/1/2024	10:30 am	10654	4
HG54587	7/1/2024	10:30 am	10456	3
HG54587	7/1/2024	10:30 am	10776	7
HG54587	7/1/2024	10:30 am	10657	5

Primary Key = (customerNumber, orderDate, orderTime, partNumber)
Foreign Key = *partNumber*

Exercise 2:

(a) Assumptions:
(1) staffNo and patNo are unique identifiers for therapist and patient respectively.
(2) An appointment is uniquely identified by (patNo, appointmentDate, appointmentTime), as a patient cannot have two appointments at the exact same time.
(3) Each appointment is with one therapist and at one branch.
(4) A therapist can work at multiple branches, but on a given day they see patients at one branch: (staffNo, appointmentDate) → branchNo.

(b) Normalizing Process:
(1) Step 1: 1NF
All values are atomic and there are no repeating groups inside a cell, so it passes 1NF.
Primary Key = **(patNo, appointmentDate, appointmentTime)**

staffNo	therapistName	**patNo**	patName	**appointmentDate**	**appointmentTime**	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

(2) Step 2: 2NF
In the table above, patName is determined only by patNo, because a patient number uniquely identifies a patient's name regardless of appointment date or time. This means patName depends only on part of the primary key, instead of on the full key (patNo, appointmentDate, appointmentTime), creating a partial dependency. So we need to split the table into new tables in reach 2NF.

patient

patNo	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

appointment

patNo	**appointmentDate**	**appointmentTime**	staffNo	therapistName	branchNo
P100	9/12/2022	10:00	S1011	Fred Smith	M15
P105	9/12/2022	12:00	S1011	Fred Smith	M15
P108	9/12/2022	10:00	S1024	Heidi Pierce	Q10
P108	9/14/2022	14:00	S1024	Heidi Pierce	Q10
P105	9/14/2022	16:30	S1032	Richard Levin	M15
P110	9/15/2022	18:00	S1032	Richard Levin	B13

Primary Key = **(patNo, appointmentDate, appointmentTime)**
Foreign Key = *patNo*

(3) Step 3: 3NF
In the above 2NF table, the primary key remains (patNo, appointmentDate, appointmentTime), while staffNo, therapistName, and branchNo are non-key attributes. However, therapistName depends on staffNo, not directly on the primary key, and branchNo depends on (staffNo, appointmentDate). Because staffNo is determined by the primary key, this creates a transitive dependency pattern: PK → staffNo → therapistName, and PK → (staffNo, appointmentDate) → branchNo. As a result, therapistName and branchNo depend indirectly on the primary key through staffNo, failing 3NF. Therefore, we need to remove these transitive dependencies to reach 3NF.

therapist

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

patient

patNo	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

therapist_day

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

appointment

patNo	**appointmentDate**	***appointmentTime***	*staffNo*
P100	9/12/2022	10:00	S1011
P105	9/12/2022	12:00	S1011
P108	9/12/2022	10:00	S1024
P108	9/14/2022	14:00	S1024
P105	9/14/2022	16:30	S1032
P110	9/15/2022	18:00	S1032

Foreign Key:
patNo → patient (patNo)
staffNo → therapist (staffNo)
(staffNo, appointmentDate) → therapist_day (staffNo, appointmentDate)

Exercise 3:

(a) Assumptions:
(1) eNo uniquely identifies an employee and determines employeeName.
(2) contractNumber uniquely identifies a contract.
(3) Each contract applies to one event, so contractNumber determines eventName.
(4) One employee can work on many contracts, and one contract can involve many employees.
(5) hoursWorked is specific to the combination of an employee and a contract.

(b) Normalizing Process:
(1) Step 1: 1NF
Each field stores a single value, and the composite primary key (eNo, contractNumber) uniquely identifies each row. Thus, the data is atomic and a primary key exists, the table passes 1NF.

eNo	**contractNo**	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

(2) Step 2: 2NF
In this table, eName depends only on eNo, since the employee number uniquely identifies the employee's name regardless of contract. Besides, because each contract applies to only one event, eventNo depends only on contractNo, not on the full key (eNo, contractNo). In addition, eventLoc depends on eventNo, and since eventNo is determined by contractNo, eventLoc is ultimately determined by contractNo rather than by the full composite key. These dependencies show that several non-key attributes are not dependent on the whole primary key, so the table fails 2NF. We need to seperate employee attributes into an employee relation and separating contract and event attributes from the employee-contract hours relationship.

employee

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

contract

contractNo	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

assignment

eNo	***contractNo***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

(3) Step 3: 3NF
In the Contract table, contractNo determines eventNo, because each contract applies to one event. Additionally, eventNo determines eventLoc, since each event has a single location. This creates a transitive dependency: contractNo → eventNo → eventLoc, which implies contractNo → eventLoc indirectly. Because eventLoc depends on contractNo through another non-key attribute (eventNo), the Contract table violates 3NF. To remove this transitive dependency, event-related attributes must be separated into an independent event (eventNo, eventLoc) relation, and the contract table should retain only contractNo and eventNo, with eventNo serving as a foreign key referencing the Event relation.

employee

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

event

eventNo	eventLoc
H25	Queens
H4	Yonkers

contract

contractNo	*eventNo*
C1024	H25
C1025	H4
C1026	H25

assignment

eNo	***contractNo***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10